

IGCSE Physics 0625 — Equation & Data Sheet

Cambridge IGCSE 0625 (2023–2027). Every equation with quantities and units, by topic.

MOTION & FORCES

Speed	$v = d/t$ v (m/s), d (m), t (s)
Acceleration	$a = (v - u) / t$ (m/s ²)
Equations of motion	$v^2 = u^2 + 2as$; $s = ut + \frac{1}{2}at^2$ (uniform a)
Force	$F = ma$ (N, kg, m/s ²)
Weight	$W = mg$ $g \approx 9.8$ N/kg (use 10 unless told)
Momentum	$p = mv$ (kg·m/s)
Impulse	$Ft = \Delta p$
Moments	moment = $F \times d$ ($d \perp$ from pivot)
Hooke's law	$F = kx$ (N/m, m)
Pressure	$p = F/A$ (Pa = N/m ²)

ENERGY & WORK

Work done	$W = Fd$ (J)
Kinetic energy	$KE = \frac{1}{2}mv^2$
GPE	$GPE = mgh$
Power	$P = W/t = E/t$ (W)
Efficiency	= useful output / total input (×100%)

THERMAL PHYSICS

Specific heat capacity	$E = mc\Delta\theta$, c (J/(kg·°C))
Specific latent heat	$E = mL$ L (J/kg)
Gas laws	$pV = \text{const}$ (isothermal); $p/T = \text{const}$ (isochoric, T in K)
Kelvin	$K = ^\circ\text{C} + 273$

WAVES & OPTICS

Wave speed	$v = f\lambda$ f (Hz), λ (m)
Refractive index	$n = \sin i / \sin r = c/v$
Critical angle	$\sin C = 1/n$
Magnification	= image height / object height

ELECTRICITY & MAGNETISM

Charge	$Q = It$ (C, A, s)
Voltage	$V = W/Q$ ($V = J/C$)
Ohm's law	$V = IR$ (⊙)
Electrical power	$P = VI = I^2R = V^2/R$
Energy	$E = VIt$
Resistance rules	series: $R = R_1 + R_2 + \dots$; parallel: $1/R = 1/R_1 + 1/R_2 + \dots$
Transformer	$V_s/V_p = N_s/N_p$ (ideal)

ATOMIC PHYSICS

Half-life	time for half the unstable nuclei to decay (graph work)
Background count	always subtract from measured count rate

UNITS & PREFIXES

Base units

m, kg, s, A, K

Prefixes

n = 10^9 , μ = 10^{-6} , m = 10^{-3} , k = 10^3 , M = 10^6 , G = 10^9

Conversions

1 m² = 10⁴ cm²; 1 m³ = 10⁶ cm³; 1 g = 10⁻³ kg